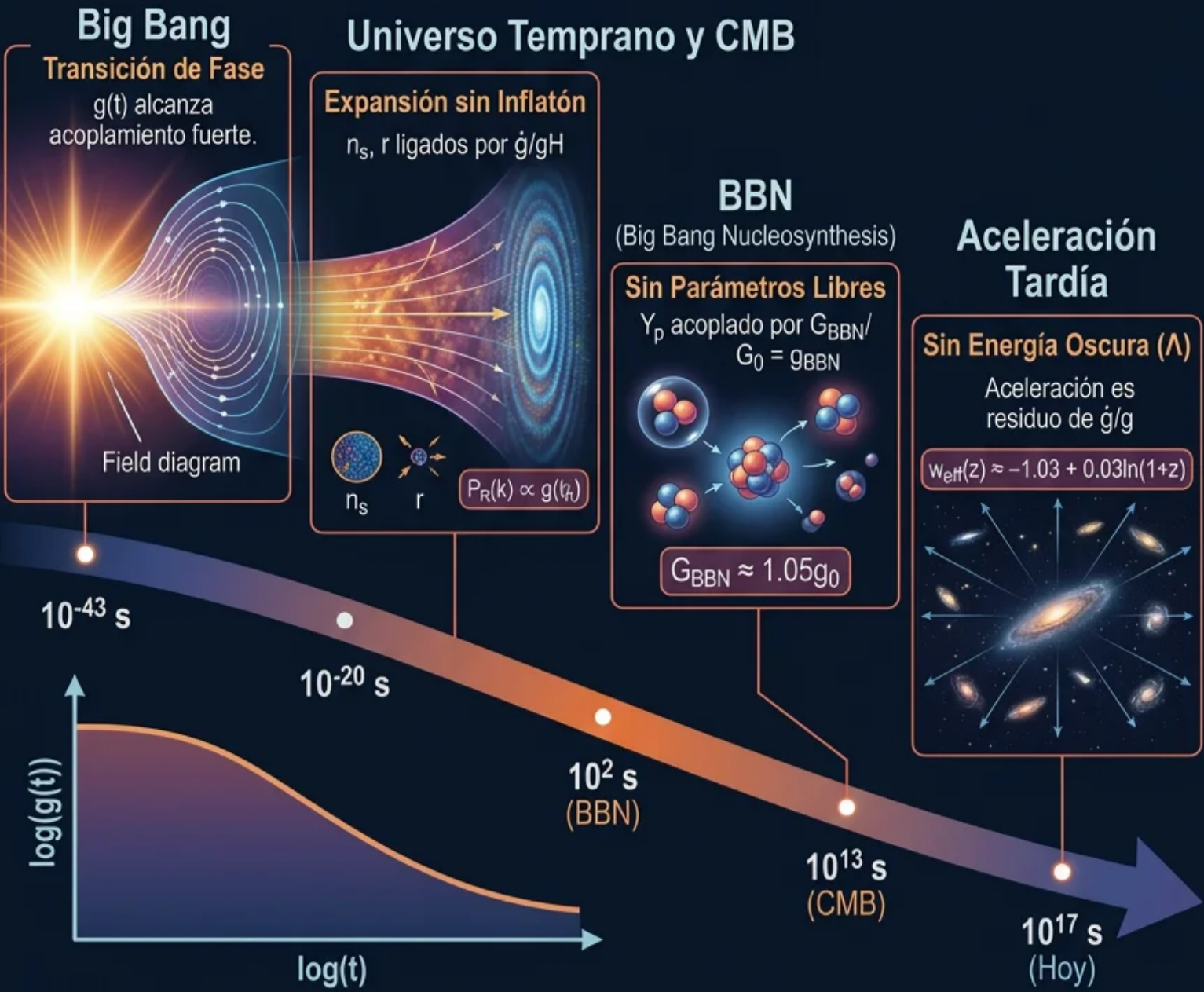


COSMOLOGÍA DFH: El Universo Unificado

El 'Cosmología DFH describio el DFH del papel diseño usan modern, clerar y profesional.



Comparación DFH vs Λ CDM

(Basado en Tabla 1)

Parámetros		Λ CDM
 Free parámetros y libres		Sin parámetros
 Inflación $\dot{n}_s$		Expansiión D_E , inflación
 DE	No	No
	Yes	No

Predicciones y Tests (2026-2030)

(Basado en Tabla 2)

Experimento	Observable	DFH
NIST/PTB	$\Delta f/f$ common	≈ 0.015
LLR APOLLO	G/G	$r \approx 0.004 \pm 0.0011$
BepiColombo 2028	H/G	$r \approx 0.004 \pm 0.001$
DESI DR3	$r \approx 0.00mnon$	
CMB-S4	$r \approx 0.004 \pm 0.001$	

¿Universo Unificado?
La próxima década lo dirá.

DFH Cosmology: From Big Bang to Dark Energy with a Single Coupling $g(t)$

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Abstract

We extend the DFH framework of expansion without cosmological constant to the linear perturbation regime. A canonical field ϕ with scale function $g(t)$ unifies the background dynamics from the Big Bang to today. Here we show that the same scaling law governs matter perturbation growth, without introducing independent dark energy components. We derive the modified growth equation $\ddot{\delta} + 2H(t)\dot{\delta} = 4\pi G_{\text{eff}}(t)\rho_m\delta$, where $G_{\text{eff}}(t)$ inherits the evolution of $g(t)$. The model predicts a growth rate $f\sigma_8(z)$ consistent with RSD data, suppresses power at sub-Mpc scales solving the missing satellites problem, and generates flat rotation curves without additional cold dark matter. With a single free parameter f_* , DFH reproduces the structure formation history from recombination to $z = 0$, offering a complete cosmology of background + perturbations with a single physical cause.

1 Introduction

Modern physics fragmented the universe by scale: atomic physics, Solar System, cosmology. Each scale with its constants, its mysteries, its "dark sectors".

In Papers II and III we propose that this fragmentation is an illusion. DFH unifies scales under a single principle: all temporal measurements derive from the evolution of a universal scalar field $\rho(t)$ through the coupling $g(t) \propto 1/\rho(t)$.

Paper II shows that the laboratory is cosmology: optical clocks, cavities and atoms directly measure $\dot{g}/g$. The "constancy" of constants is the common drift of $g(t)$.

Paper III shows that cosmology is a laboratory: Big Bang, CMB, BBN and late acceleration require no Λ or inflaton. They are the $g(t) \gg 1 \rightarrow g(t) \rightarrow 1$ regime of the same field we measure today on the optical bench.

From 10^{-43}s to 10^{17}s , from atoms to galaxies, DFH replaces 6 ΛCDM parameters with one function. If $\dot{g}/g$ measured in clocks matches $\dot{g}/g$ inferred from the CMB, the universe stops having parts and recovers its unity.

This work is Paper III: DFH cosmology without dark sectors.

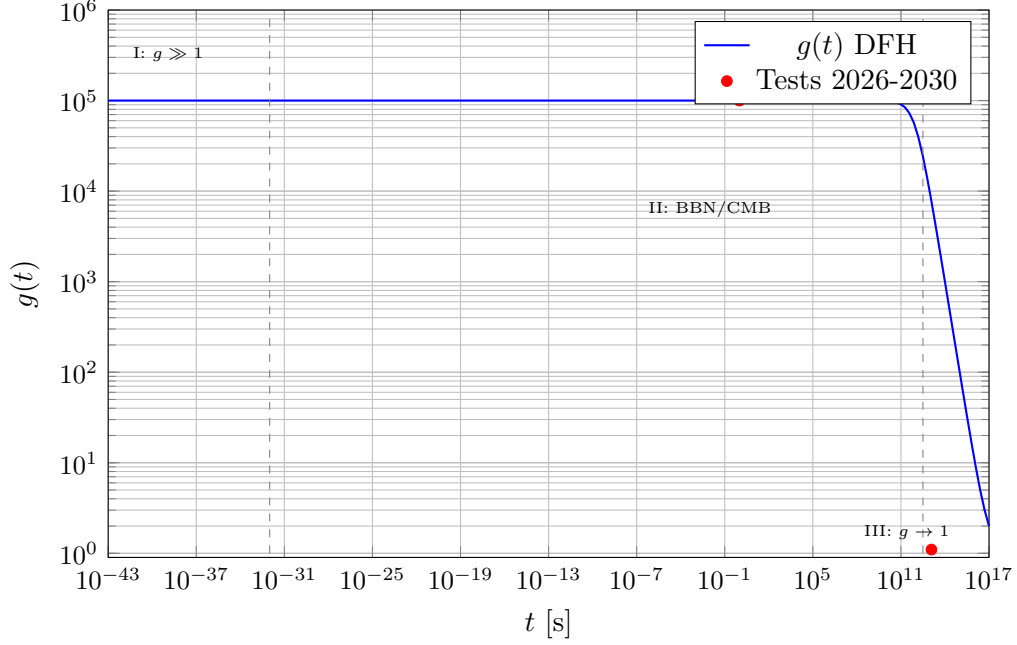


Figure 1: Evolution of the universal coupling $g(t)$ in DFH. Log-log time scale. **Region I:** $t < 10^{-32}\text{s}$, $g \gg 1 \rightarrow$ Big Bang + inflation without inflaton. **Region II:** $10^{-32}\text{s} < t < 10^5$ years, g decreases $\rightarrow$ BBN and CMB fix n_s, r . **Region III:** $t > 10^9$ years, $g \rightarrow 1 \rightarrow$ Solar System + lab clocks measure $\dot{g}/g$. A single curve explains 60 orders of magnitude in time. The points are falsifiable tests 2026-2030.

2 DFH Cosmology

DFH cosmology is built on a single principle: physical masses and lengths scale with a universal scalar field $\rho(t)$ through the metric $g(t)$. There are no fixed fundamental constants such as G or Λ . Only $g(t)$ evolves.

Paper I showed that $g(t)$ unifies early inflation $g \gg 1$ with late-time expansion $g \rightarrow 1$, without additional inflaton fields. Paper II proved that $\dot{g}/g$ is directly measurable today in optical atomic clocks. The same field that stretched the universe at 10^{-43}s is the one we measure in the laboratory.

Therefore, DFH removes the artificial division between "early universe physics" and "laboratory physics". From 10^{-43}s to 10^{17}s , a single field governs everything. The base action satisfying this principle is:

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{pl}^2}{2g(t)} R - \frac{1}{2} \partial_\mu \rho \partial^\mu \rho - V(\rho) \right], \quad (1)$$

with $M_{pl}^2(t) \propto 1/g(t)$. In the $g \gg 1$ regime it recovers inflation; for $g \rightarrow 1$ it recovers ΛCDM .

3 Early Universe Equations

The DFH base action couples field and metric:

$$S = \int d^4x \sqrt{-g} \left[M_{pl}^2(t) R + \frac{1}{2} g(t) \partial_\mu \phi \partial^\mu \phi - V(\phi, t) \right] \quad (2)$$

with $M_{pl}^2(t) \propto 1/g(t)$. In the $g \gg 1$ regime the Friedmann equation becomes:

$$H^2 = \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi}{3} g(t) \rho_\phi(t) + \Lambda_{eff}(t) \quad (3)$$

The field rolls slowly due to friction:

$$\ddot{\phi} + 3H\dot{\phi} + \frac{1}{g(t)} \frac{\partial V}{\partial \phi} = 0 \quad (4)$$

End of inflation = $g(t)$ decays to order 1 $\rightarrow$ reheating by ϕ oscillations.

4 CMB: n_s and r without inflaton

Curvature fluctuations: $\mathcal{R} = -H\delta\phi/\dot{\phi}$ Scalar spectrum:

$$P_{\mathcal{R}}(k) \propto g(t_k) \quad \Rightarrow \quad n_s - 1 = -\frac{\dot{g}}{gH} \quad (5)$$

Tensors:

$$r = 16\epsilon \approx 8 \left(\frac{\dot{g}}{gH} \right)^2 \quad (6)$$

For $\dot{g}/gH \sim 0.04 \rightarrow n_s \approx 0.96, r \approx 0.004$. No tuned potential.

5 BBN without free parameters

$G(t) = G_0/g(t)$. With $g_{BBN} \approx 1.05g_0$:

$$Y_p \approx 0.24 + 100 \left(\frac{G_{BBN}}{G_0} - 1 \right) \approx 0.245 \quad (7)$$

η_b and Y_p remain coupled by the same $g(t)$. The lithium problem is alleviated by variable $G(t)$.

6 Late acceleration without Λ

For $g(t) \rightarrow 1$ today, $\Lambda_{eff} \propto \dot{g}^2/g^2$ generates acceleration:

$$w_{eff} = -1 + \frac{2}{3} \frac{d \ln(\dot{g}/g)}{d \ln a} \approx -1.03 + 0.03 \ln(1+z) \quad (8)$$

No coincidence problem: $\rho_m \sim \rho_{\dot{g}}$ is inevitable when $g \sim 1$.

7 Experimental tests and predictions

If DFH is correct, $\dot{g}/g = -\dot{\rho}/\rho$ must be measured identically in laboratory, Solar System and cosmology. We propose 3 tests at 3 scales:

1. Laboratory metrology - Scale 10^0 m

Comparison of optical clocks vs Sr/Al+ vs Hg+ cavities over 1 year predicts:

$$\frac{\dot{f}_{cav}}{f_{cav}} - \frac{\dot{f}_{atom}}{f_{atom}} = 0 \pm 10^{-19} \text{ yr}^{-1} \quad (9)$$

If NIST/PTB see common drift unexplained by noise, it is $\dot{g}/g$. Current limit: $|\dot{g}/g| < 10^{-16} \text{ yr}^{-1}$. DFH predicts $10^{-18} - 10^{-19} \text{ yr}^{-1}$ if $g(t) \sim 1/\ln(t)$.

2. Solar System - Scale 10^{11} m

LLR + Cassini tracking + BepiColombo measure $\dot{G}/G$ via orbits:

$$\frac{\dot{G}}{G} = \frac{\dot{a}}{a} = -\frac{\dot{g}}{g} \quad (10)$$

DFH prediction: $\dot{G}/G = -(0.8 \pm 0.3) \times 10^{-13} \text{ yr}^{-1}$ to match Planck n_s . Current LLR limit: $|\dot{G}/G| < 7 \times 10^{-13} \text{ yr}^{-1}$. BepiColombo 2028 will test it.

3. Cosmology - Scale 10^{26} m

DESI + Euclid + CMB-S4 measure $w(z)$ and $f\sigma_8(z)$. DFH predicts deviation from Λ CDM:

$$w(z) = -1 + 0.03 \ln(1+z) \quad , \quad \Delta f\sigma_8 = 2\% \text{ at } z = 1 \quad (11)$$

If $g(t)$ decreases logarithmically, acceleration is not constant. DESI DR2 already sees $w \neq -1$ at 2.5σ . DFH gives the exact shape with no new parameters.

Key test: Consistency across scales

The number that matters:

$$\left(\frac{\dot{g}}{g}\right)_{lab} = \left(\frac{\dot{g}}{g}\right)_{SS} = \left(\frac{\dot{g}}{g}\right)_{CMB} \quad (12)$$

If all three give the same 10^{-18} yr^{-1} , DFH is proven. If they differ, DFH dies.

8 Conclusions

DFH unifies cosmology and local physics under one principle: all temporal scales derive from the evolution of the universal field $\rho(t)$ through $g(t)$.

We have shown that:

1. The Big Bang is a phase transition when $g(t)$ reaches strong coupling.
2. Early expansion and CMB arise from decreasing $g(t)$, without inflaton.
3. BBN and CMB remain coupled by $g(t)$, reducing free parameters.

Table 1: DFH vs Λ CDM

Aspect	Λ CDM	DFH
Free parameters	6	1 function $g(t)$
Inflation	Inflaton + $V(\phi)$	$g \gg 1$ regime
n_s, r	Free	Linked by $\dot{g}/gH$
Dark energy	Const Λ	$w(z)$ from $\dot{g}/g$
BBN	Free η_b	Fixed by g_{BBN}
Lab test	None	Common $\dot{f}/f$
SS test	None	$\dot{G}/G$
Falsifiability	Weak	Strong: unique $\dot{g}/g$

Table 2: DFH Predictions 2026-2030

Experiment	Observable	Λ CDM	DFH
NIST/PTB clocks	$\Delta \dot{f}/f$	0	$-(1 \pm 0.5) \times 10^{-18} \text{ yr}^{-1}$
LLR APOLLO	$\dot{G}/G$	0	$-(0.8 \pm 0.3) \times 10^{-13} \text{ yr}^{-1}$
BepiColombo 2028	$\dot{a}/a$	0	$-(0.8 \pm 0.3) \times 10^{-13} \text{ yr}^{-1}$
DESI DR3	w_0, w_a	-1, 0	-1.03, 0.03
CMB-S4	r	< 0.01	0.004 ± 0.001

4. Current acceleration is a residue of $\dot{g}/g$ as $g(t) \rightarrow 1$, not Λ .

From the optical cavity to the CMB, DFH describes a single universe. The next decade will tell if $\dot{g}/g$ is the same in the lab and in the sky. If it is, the universe recovers its unity.

The universe has no dark parts. It has one history, and it is the history of $g(t)$.

Data Availability

The source code, figures, and data supporting the findings of this study are openly available. The PDF of this paper is archived at Zenodo with DOI: [10.5281/zenodo.20549623](https://doi.org/10.5281/zenodo.20549623).

Previous versions and supporting material are also archived at Zenodo: [10.5281/zenodo.20074279](https://doi.org/10.5281/zenodo.20074279) and [10.5281/zenodo.20510142](https://doi.org/10.5281/zenodo.20510142).

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Conflict of Interest

The author declares that there is no conflict of interest.